

CENTIME: A Direct Comprehensive Traffic Features Extraction for Encrypted Traffic Classification

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Abstract—With the rapid development of the network, encrypted traffic classification plays a vital role in guaranteeing the quality of network services and ensuring the security of the network. Recent studies show that machine learning approaches based on statistical features and raw traffic sessions are effective for this task. However, the performance of the statistical-based approaches largely depends on the quality of the features. Experts need to design different features for different encrypted traffic classification tasks, which is time-consuming. Meanwhile, the raw traffic-based approach needs to uniformize the traffic size; this will cause the loss of information about the overall structure of the network traffic; for example, we do not know the time from the first packet to the last packet in a session. This paper proposes the CENTIME, which can extract comprehensive information based on ResNet and AutoEncoder to identify encrypted traffic. ResNet is used to extract information from uniformized traffic, and AutoEncoder is used to encode statistical features. The statistical features are used to compensate for the information loss caused by traffic uniformization. They only need to be designed once rather than be designed separately for different tasks. Moreover, the pooling layers are removed, and 1D convolution layers are used to help CENTIME make more effective use of raw traffic information. We evaluate the CENTIME on the public dataset “ISCX VPN-nonVPN”, and the results demonstrate the CENTIME outperforms the state-of-the-art encrypted traffic classification methods. More importantly, comprehensive traffic features generated in the CENTIME can represent different classes of traffic well.

Index Terms—Encrypted traffic classification, ResNet, AutoEncoder

I. INTRODUCTION

With the rapid development of the Internet, network traffic from different applications has risen rapidly. A large amount of traffic makes it difficult for network operators to guarantee the quality of services (QoS) [1] and ensure Internet security [2]. Traffic classification, as one of the necessary technologies for improving network service and ensuring network security,

has received increased attention from academic and industrial fields these years [3].

The fastest and most effective approach to identify traffic is to use the port number [4]. Some well-known Internet services use the fixed port number, such as the HTTP service applications usually use port 80 while FTP service applications use port 21. However, as more and more new applications avoid using standard registered port numbers, the accuracy of the port-based approach has declined [5]. The next technology to classify traffic relies on the payload. This technology is also known as deep packet inspection (DPI) [6], which focuses on finding patterns of specific string in traffic packet. In recent years, as people have gradually paid attention to their privacy, more and more applications have started to use encrypted communication when transmitting any data, which has brought considerable challenges to the above two traditional traffic classification methods.

To solve these challenges, researchers adopt machine learning and deep learning methods for encrypted traffic classification. According to different inputs, the studies can be divided into two categories, (1) statistical characteristics-based approach and (2) raw traffic-based approach. To be specific, the statistical-based approach assumes that different types of traffic have unique statistical characteristics. Alshammari *et al.* used packet-related features like “packet length”, “delta time from previously captured packet”, and “IP flags” to identify the encrypted traffic [7]. The raw traffic-based approach uses the deep neural network to extract features from network traffic automatically. For example, Wang *et al.* proposed an end-to-end method based on the one-dimensional convolutional neural network [8]. This method integrates feature extraction and classifier into one unified framework. The result shows that this method can achieve state-of-the-art performance on the “ISCX VPN-nonVPN” dataset [9].

Although the previous technologies can obtain comparatively high accuracy, there still exists apparent shortcomings of them. For the statistical-based approach, the performance

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of the classification largely depends on the quality of the features. Experts need to design different features for different encrypted traffic classification tasks, which is time-consuming and expensive. The main drawback of the raw traffic-based approach is that it will cause a loss of information. Most studies use convolutional neural networks to extract features from raw encrypted traffic [8], [10], [11]. However, the convolutional neural networks need a fixed-sized input; therefore, the raw encrypted traffic needs to be uniformized first. This will cause information loss, especially information about the overall structure of the network traffic. For example, we cannot know the number of packets in a session, the time from the first packet to the last packet in a session, and the minimum or maximum delta time in a session after uniformizing, which will reduce the classification performance.

For the sake of solving these limitations, we proposed a framework named CENTIME. CENTIME can extract comprehensive features from network traffic and use these features to classify encrypted traffic. There are three parts in CENTIME, namely data preprocessing, traffic information extraction and traffic classification. (1) In the part of data preprocessing, we split the raw traffic by session-packets and then anonymize them. (2) During the part of traffic information extraction, we use ResNet [12] to extract features directly from raw encrypted traffic. Compared with the previous researches, we removed the pooling layer and used the one-dimensional convolution layer in CENTIME, hoping to extract more valuable information from the raw traffic. Simultaneously, statistical information is extracted from raw traffic too. We use AutoEncoder [13] to compress statistical information. This statistical information is used to compensate for the information loss caused by traffic trimming. They only need to be designed once, containing information about the overall structure of the traffic, rather than be designed separately for different tasks. Finally, the information extracted by ResNet and encoded by AutoEncoder are merged to form the comprehensive features. (3) The comprehensive features are used to classify the encrypted traffic in the last part of CENTIME. To demonstrate the feasibility of this framework, we perform 21 different experiments, compared the CENTIME with six state-of-the-art methods on the “ISCX VPN-nonVPN” dataset [9]. The final results show that our proposed framework outperforms other public methods in encrypted traffic classification.

The major contributions of this paper are as follows:

- We propose a novel framework named CENTIME, which integrates traffic information extraction and encrypted traffic classification into one framework. The framework generates comprehensive features based on the uniformized traffic files and statistical information, then uses these comprehensive features to classify encrypted traffic.
- In this framework, we use 1D convolution layers and remove the pooling layers to make more effective use of raw traffic information.
- We evaluate our model with the six state-of-the-art methods on the public dataset “ISCX VPN-nonVPN”.

Experimental results show that our method has a good performance on classification; it achieves an f1-score of 0.99 in encrypted traffic classification. Besides, all the relevant experimental data, such as training data, the code for training, fine-tuned models and detailed experiment results are published on GitHub¹.

The rest of the paper is organized as follows. Section II summarizes the related work on traffic classification. Section III describes the CENTIME framework, including three parts, data preprocess, traffic information extraction and traffic classification. Section IV covers the experiment setup, evaluation metrics, baseline methods and experiments results. Finally, section V provides the conclusion and future studies that can be engaged in.

II. RELATED WORK

Traffic classification technologies have developed fast over time. There have been two traditional methods for classifying network traffic: “port-based” and “content-based” [5]. Among all these mature methods, deep packet inspection (DPI) is considered one of the most reliable method [6]. DPI will not only use the information in the packet headers (such as the port number), but also use the payload of the traffic to classify the network traffic. More specifically, DPI uses a signature database containing information from some labelled packets to identify the traffic. However, with the increased use of HTTPS and other encryption methods like VPNs, the classification accuracy based on DPI gradually decreases. We cannot decrypt the payload; therefore, we cannot classify the network traffic by using pattern matching to search for particular keywords in the payload.

In order to classify encrypted traffic, the statistical-based method is proposed. Most of the research based on traffic statistical features uses traditional machine learning algorithms, such as decision tree or support vector machine (SVM). For example, [14] presents a statistical classifier based on a combination of k-nearest neighbour (K-NN) and k-means that allows real-time classification of encrypted traffic. Alshammari *et al.* use C4.5 [15] and AdaBoost [16] to unveil encrypted application based on statistical flow feature [7]. Agrawal *et al.* use five supervised ML algorithms, such as Naive Bayes Tree and NaiveBayes, to identify P2P traffic based on statistical features like the port number [17]. Alshammari *et al.* also employ statistical network traffic flow feature to identify encrypted VoIP application in network traffic [18]. Vlăduțu *et al.* propose a new technique by combining unsupervised algorithms (K-means, Expectation maximization) and supervised algorithms (Decision tree) to classify Internet traffic [19]. Vu *et al.* generate fake network traffic with the Synthetic Minority Over-sampling Technique (SMOTE) [20] and modify the cost function to address the imbalanced encrypted traffic classification problem based on statistical characteristics [21]. However, the performance of these models heavily depends on the quality

¹<https://github.com/wmn7/Traffic-Classification>

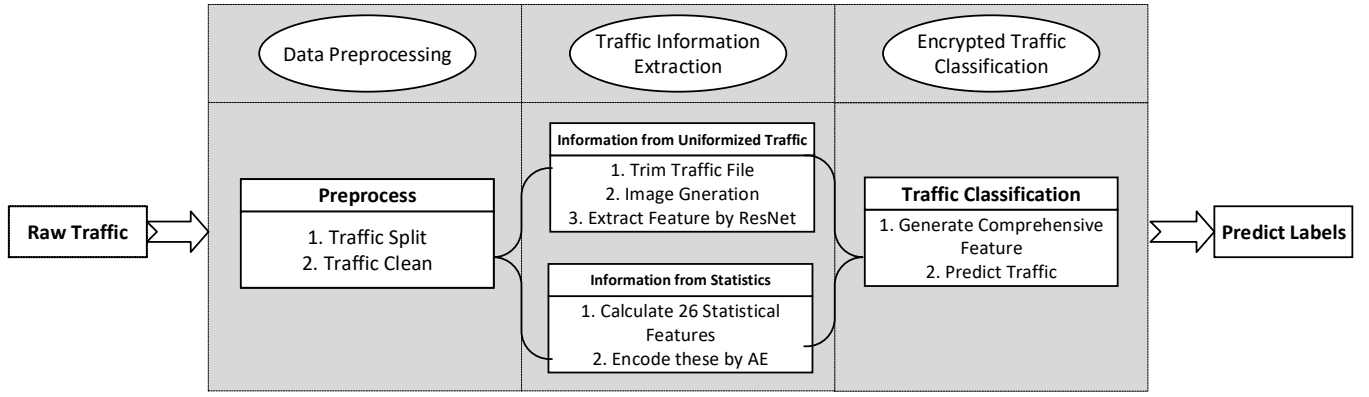


Fig. 1: Overview of the Structure of CENTIME.

of the human-engineered features, which significantly limits the generalizability of the statistical-based method.

Recently, deep learning methods have been widely used in encrypted traffic classification. The neural networks can automatically extract features from the traffic payload, increasing the generalization of the model. Wang *et al.* propose an end-to-end framework based on the convolution neural network, which combines feature extraction, feature selection, and classifier into one system [8]. Song *et al.* used a text convolutional neural network to extract critical features for traffic classification [10]. [22] proposes a session-packets-based encrypted network traffic classification model using capsule neural networks (CapsNet), which can learn the spatial characteristics from network traffic well [23]. Shapira *et al.* transformed the basic flow data into a picture and then used image classification deep learning techniques to classify encrypted traffic [24]. Lotfollahi *et al.* proposed a scheme called “Deep Packet”, which uses 1D-CNN and stacked autoencoder (SAE) to extract features and make predictions [25].

III. THE PROPOSED METHOD

In this section, we propose a framework named CENTIME, which is based on ResNet and AutoEncoder. As shown in “Fig. 1”, CENTIME consists of three parts: data preprocessing, traffic information extraction and traffic classification. During the data preprocessing phase, we split raw traffic by session-packets, and then delete MAC and IP address. During the phase of traffic information extraction, 26 statistical features are calculated from raw traffic. After that, we uniformize the raw traffic and then use ResNet to extract information from the traffic. During the classification phase, we combine the information from uniformized traffic and statistical features to generate comprehensive features and use these features to classify encrypted traffic. Dataset is also introduced in this section.

A. Dataset

The most critical thing in deep learning is to have a large number of datasets to avoid overfitting. However, there are relatively few datasets in the field of encrypted traffic

TABLE I: Introduction to 12 Classes of Traffic in the “ISCX VPN-nonVPN” Dataset

Non-VPN/VPN	Traffic Type	Total Size
non-VPN	Email	13 MB
	Chat	29.5 MB
	Streaming	1.53 GB
	File Transfer	17.3 GB
	P2P	96.8 MB
	VoIP	4.48 GB
VPN	VPN-Email	7.8 MB
	VPN-Chat	27.6 MB
	VPN-Streaming	1.37 GB
	VPN-File Transfer	279 MB
	VPN-P2P	358 MB
	VPN-VoIP	360 MB

classification. There are only some datasets generated in the process of encrypted traffic identification research. For example, UMass² provides datasets related to Tor traffic [26] and HTTPS traffic [27]. The “Campus” dataset is used to identify different encryption algorithms [28].

Draper-Gil *et al.* proposed the “ISCX VPN-nonVPN” dataset in 2016 [9]. There are 14 classes of traffic files, which captured from different applications (i.e., Skype, BitTorrent), in the “ISCX VPN-non VPN” dataset, including seven classes of regular (non-VPN) traffic and seven classes of VPN traffic. Because the “ISCX VPN-nonVPN” dataset includes rich types of applications, it is widely used in the encrypted traffic research field. The experiments in this paper are also based on the “ISCX VPN-nonVPN” dataset.

In the “ISCX VPN-nonVPN” dataset, the authors do not give the labels to the traffic files but only give each traffic file a brief description. This leads to some ambiguous traffic files.

²<http://traces.cs.umass.edu/index.php/Network/Network>

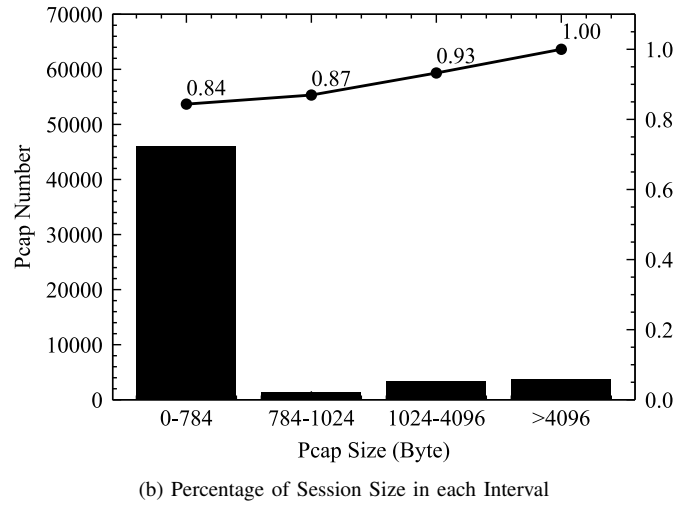
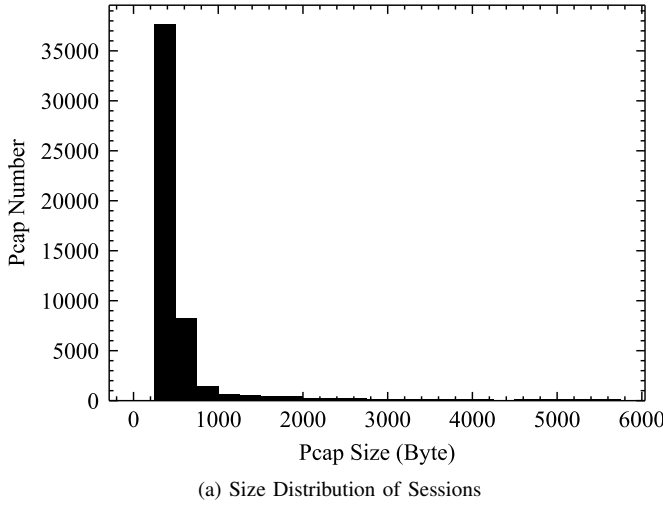


Fig. 2: Analysis of Session Size

For example, “Facebook_video.pcap” can be either belong to “Browser” or “Streaming”. Therefore, we removed these ambiguous traffic files in this paper, and the final “ISCX VPN-nonVPN” dataset has 12 classes. “Table. I” introduces the detailed content of these 12 classes.

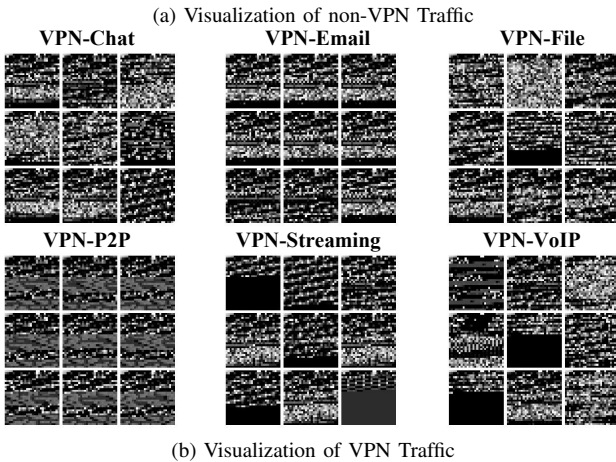
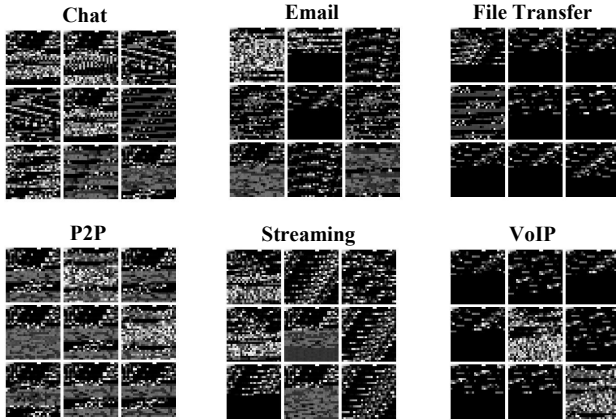


Fig. 3: Visualization of Encrypted Traffic (784 bytes)

B. Data Pre-processing

Before extracting information from the traffic, we need to perform data preprocessing first. There are two steps of data preprocessing in the CENTIME framework, traffic split and traffic clean.

Traffic Split. In this step, we divide the continuous traffic into multiply discrete traffic units. The continuous raw traffic P contains the different size of packets p^i , that is $P = \{p^1, p^2, \dots, p^N\}$. N represents the number of packets in raw traffic P . A single packet p^i is defined as (1),

$$p^i = (x^i, t^i, b^i), \quad (1)$$

where x^i is the five-tuple, source IP, source port, destination IP, destination port, and transport-level protocol. t^i is the start time of the packet, and b^i is the byte length of the packet.

Session and flow are two common ways to split the traffic. A flow P_{flow} is a group of traffic that has the same five-tuple, which can be defined as (2).

$$P_{flow} = \{p^1 = (x^1, t^1, b^1), \dots, p^n = (x^n, t^n, b^n)\}, \quad (2)$$

where $x^1 = \dots = x^n$ and $t^1 < \dots < t^n$. The session is very similar to flow, but the source IP and destination IP can be interchanged [29]. We split the traffic into session in this paper because Wang *et al.* found that the session contains more interaction information than flow and is more suitable for encrypted traffic classification [8].

Traffic Clean. After splitting the traffic into the session, we delete their IP address and MAC address by masking related strings with 0x00. Each traffic class has a unique IP address in the “ISCX VPN-nonVPN” dataset; therefore, the model may overfit if the IP address and MAC address are not deleted. Empty files and duplicate files are also removed during traffic clean.

C. Traffic Information Extraction

Our framework CENTIME can generate comprehensive traffic features from the raw traffic. The comprehensive features are composed of two parts: the statistical information of the raw traffic and the uniformized traffic information.

Information from Statistics (IFS). Before uniformizing the traffic, we extract statistical features from it firstly. These features can retain information on the overall structure of the traffic. In this paper, we extract 26 statistical features, such as “Num pkts” (the number of packets in a session), “Avg syn flag” (the average of packets of with syn flag active in a session), and “Duration window flow” (the time from the first packet to the last packet in a session), etc. The detailed information of these 26 extracted features can be seen in [30]. After calculating the 26 statistical features, the min-max normalization method is used to normalize data. That is,

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}. \quad (3)$$

After the normalization process, all statistical features are converted to a range $[0, 1]$. The AutoEncoder is used to encode these 26 statistical features and extract effective information from them.

Information from Uniformized Traffic (INUIT). In this step, we use ResNet to extract information directly from the raw traffic. Using ResNet needs a fixed-sized input, so we uniformize the raw traffic to the same size. Suppose we uniformize the raw traffic S_i into N bytes. If the size of S_i is larger than N bytes, only the first N bytes are retained; otherwise, 0x00 will be added at the end of the traffic.

“Fig. 2a” shows the size distribution of the session in the “ISCX VPN-nonVPN” dataset. It can be seen that most of the session size is less than 1000 bytes. In this paper, the raw traffic is uniformized into 784 bytes, 1024 bytes, and 4096 bytes ($N = 784, 1024, 4096$), respectively. All the experiments in this paper are based on these three types of traffic data. As shown in “Fig. 2b”, the session size in the range of $[0, 784]$ bytes accounted for 84% of the traffic, while only 3% of the session size was in the range of $[784, 1024]$ bytes. A total of 93% of the session size is between $[0, 4096]$ bytes.

After all the traffic is uniformized, each byte in the session corresponds to the grey-scale pixel value; for example, 0x00 means black, and 0xff means white. As shown in “Fig. 3”, the 784 bytes session can be converted into a 28×28 matrix. “Fig. 3a” shows the visualization of all non-VPN traffic classes, and “Fig. 3b” shows the visualization of the VPN traffic. It can be seen that patterns are different for different traffic classes, while among the same class of traffic, their patterns are similar. Therefore, there is reason to presume that ResNet can extract valid information from uniformized traffic.

D. Encrypted Traffic Classification (ERIC)

After the “Traffic Information Extraction” process, we got the information from two parts: the information obtained from the statistical features through the AutoEncoder $V_{statistics}$, and the uniformized traffic information extracted by the

ResNet V_{raw_pcap} . The two parts of information are merged to form comprehensive traffic features V . We believe that the comprehensive traffic features V include information both from uniformized traffic and statistics of traffic. Then we use the features V to classify encrypted traffic by using a fully connected neural network. “Fig. 4a” shows the whole process of “Traffic Information Extraction” and “Encrypted Traffic Classification”. ResNet is used to extract information from uniformized traffic. As shown in “Fig. 4a”, there are four ResLayers in the ResNet; each ResLayer contains two ResBlock. The structure of ResBlock can be seen in “Fig. 4b”. The output channel of the convolution layer in the same ResLayer is the same.

The main parameters of the models in CENTIME are described in “Table. II”. There are two points to note in the CENTIME framework, (1) 1D convolution layer is used because the traffic data is sequence data. The performance of the 1D convolution layer and 2D convolution layer will be compared in section IV. (2) The pooling layer is removed in the framework. The function of the pooling layer is to reduce the spatial size of the input to reduce the number of parameters in the network. However, the uniformized traffic size is not large in this experiment, so we removed the pooling layer to retain more helpful information. The performance of the model with the pooling layer and without the pooling layer will also be compared in section IV.

TABLE II: Main Parameters of the CENTIME

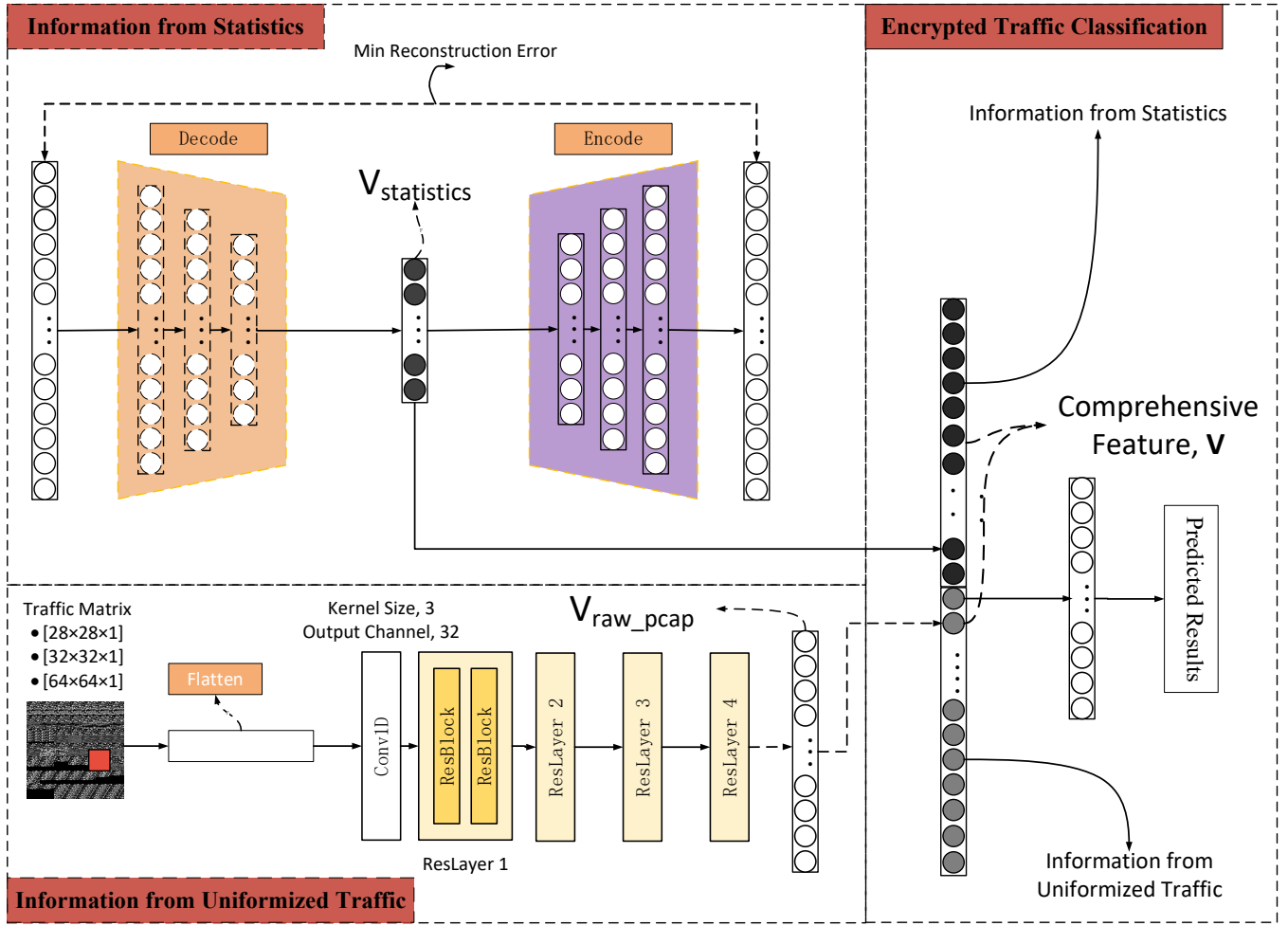
	Layer	Operation	Input	Filter	Output
INUIT	Conv-1	conv1d * 1	1*784	1*9	32*784
	ResLayer-1	conv1d * 4	32*784	1*3	32*784
	ResLayer-2	conv1d * 4	32*784	1*3	64*364
	ResLayer-3	conv1d * 4	64*364	1*3	128*182
	ResLayer-4	conv1d * 4	128*182	1*3	256*91
	Avg-Pooling Flatten	Avg Pooling flatten	256*91 256*1	- -	256*1 256
IFS	Auto-Encoder-1	fully-connected	26	-	18
	Auto-Encoder-2	fully-connected	18	-	9
	Auto-Encoder-3	fully-connected	9	-	18
	AutoEncoder-4	fully-connected	18	-	26
ERIC	Fully-connected-1	fully-connected	256+9	-	100
	Fully-connected-2	fully-connected	100	-	30
	Fully-connected-3	softmax	30	-	12

IV. EXPERIMENT

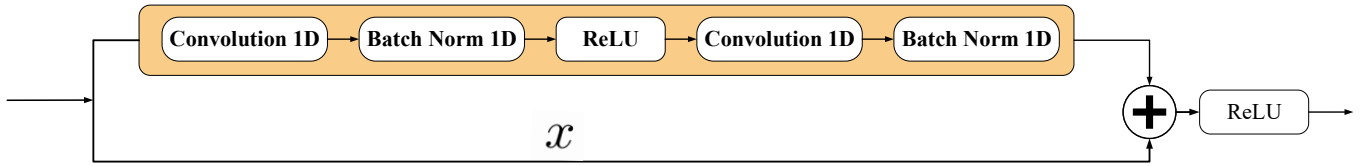
A. Experiment Setup

All the experiments are conducted on Ubuntu 16.04 64bit OS. The hardware of the sever is Inter (R) Xeon (R) E5-2650 2.2 GHz with 32GB of memory. An Nvidia GTX1080Ti 11GB GPU is used as the accelerator.

We use PyTorch [31] to build all the neural networks in this paper. We randomly sampled 90% of the data as the training data, and the rest is the test data. The mini-batch size is 256. The initial learning rate is 0.001 and decay of 0.9 per 25 epochs. Adam optimizer [32], which inherits from RMSProp



(a) The Process of "Traffic Information Extraction" and "Encrypted Traffic Classification"



(b) Structure of ResBlock

Fig. 4: The Architecture of the Encrypted Traffic Classification Model

and AdaGrad [33], is used to train the deep neural networks in this paper. The total training time is 150 epochs in each experiment. All the detailed configurations can be found in "traffic_classification.yaml" on our shared Github repository¹.

B. Baseline Models

Three baseline methods are included to verify the performance of CENTIME.

CNN1D, CNN1D method uses 1D convolution layers to extract features from the traffic files directly and then use these features to classify encrypted traffic.

CNN2D, CNN2D method is very similar to CNN1D, except that 2D convolution layers are used for feature extraction. The

network structure of CNN1D and CNN2D can be seen in [8]. At the same time, the above two baseline methods will include two forms, with and without pooling layers.

ResNet, the third baseline method uses ResNet to extract information from network traffic. The structure of ResNet here is the same as the structure in "Information from Uniformized Traffic" in CENTIME.

More detailed information on the above three baseline models can be referred to the GitHub¹.

C. Evaluation Metric

In our experiment, four evaluation metrics are used to evaluate the performance of our model, that is, accuracy

TABLE III: Result of CENTIME Compared with Baseline Method for Encrypted Traffic Classification

	784 Bytes				1024 Bytes				4096 Bytes			
	Acc	Pr	Re	F_1	Acc	Pr	Re	F_1	Acc	Pr	Re	F_1
CNN1D-Pooling	0.9398	0.9397	0.9398	0.9397	0.9523	0.9522	0.9523	0.9521	0.9525	0.9529	0.9525	0.9526
CNN1D-noPooling	0.9778	0.9779	0.9778	0.9778	0.9765	0.9766	0.9765	0.9765	0.9697	0.9696	0.9697	0.9696
CNN2D-Pooling	0.9313	0.9312	0.9313	0.9312	0.9366	0.9364	0.9366	0.9365	0.9161	0.9169	0.9161	0.9162
CNN2D-noPooling	0.9347	0.9350	0.9347	0.9348	0.9612	0.9612	0.9612	0.9611	0.9301	0.9309	0.9301	0.9303
ResNet1D	0.9913	0.9914	0.9913	0.9913	0.9909	0.9910	0.9909	0.9909	0.9934	0.9935	0.9934	0.9934
ResNet2D	0.9770	0.9772	0.9770	0.9771	0.9742	0.9741	0.9742	0.9741	0.9717	0.9718	0.9717	0.9716
CENTIME	0.9979	0.9981	0.9979	0.9980	0.9972	0.9973	0.9972	0.9973	0.9977	0.9979	0.9977	0.9979

(Acc), precision (Pr), recall (Re), and F1-score (F_1). Accuracy (Acc $\in [0, 1]$) means the proportion of correct predictions among the total number of the dataset. We use accuracy to evaluate the overall performance of the model. Precision (Pr $\in [0, 1]$) means the percentage of the relevant results. Recall (Re $\in [0, 1]$) means the percentage of total results correctly classified by the model. F-measure ($F_1 \in [0, 1]$) can measure precision and recall at the same time. We calculate the harmonic mean of precision and recall to get the F_1 . F_1 can be defined as (4),

$$F_1 = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

D. Experiment Results and Analysis

To verify the effectiveness and performance of CENTIME, we perform 21 experiments with different models and different uniformized sizes for encrypted traffic classification task on the “ISCX VPN-nonVPN” dataset. The experimental results, including accuracy, precision, recall and average F-measure, are shown in the following “Table. III”. These 21 experiments are mainly used to illustrate the following three problems, (1) the impact of the different uniformized sizes of the traffic files on the final result. (2) the impact of the pooling layer on the final performance. (3) comparing the effects of 1D convolution layers and 2D convolution layers on the results.

Comparison of Different Uniformized Sizes: In section III-C, all traffic files are uniformized into the same sizes. We use different uniformized sizes of traffic to classify encrypted traffic and compare the final results. As shown in “Table. III”, only the performances on “CNN1D-Pooling” and “CNN2D-Pooling” improve when the traffic size increases from 784 bytes to 1024 bytes, while the performances of other models have decreased. Similarly, when the traffic size increases from 1024 bytes to 4096 bytes, the accuracy of most models decreases too. This result is reasonable. As shown in “Fig. 2”, most of the session sizes are around 784 bytes. If the raw traffic is uniformized to 4096 bytes, a large amount of 0x00 needs to be added at the end of the traffic file. Therefore, the data will contain much invalid information, which can decrease the performance of the models.

Comparison of With and Without Pooling Layer: As shown in “Table. III”, from “CNN1D-Pooling” to “CNN1D-noPooling”, the accuracy of the model has increased significantly. For example, when the traffic size is 784 bytes,

the accuracy increases 4% when removing the pooling layer. Similarly, in the case of 2D models (like “CNN2D-Pooling” and “CNN2D-noPooling”), the performances of models are also improved significantly when not using pooling layers. The main reason is that when removing the pooling layer, the raw traffic can retain more helpful information and help the model make a judgment.

Comparison of 1D Convolution Layer and 2D Convolution Layer: It can be seen from “Table. III” that the accuracy of the models with the 1D convolution layer is higher than the models with the 2D convolution layer. For example, the accuracy of “ResNet1D” is 2% higher than “ResNet2D” when the session size is 784 bytes. It proves that the 1D convolution layer is more suitable for encrypted traffic classification than the 2D convolution layer. This conclusion is similar to Wang’s work [8].

In summary, only increasing the traffic file size cannot significantly improve the performance of the models. In this paper, it is sufficient to uniformize the traffic size to 784 bytes. However, removing the pooling layer and using 1D convolution layer can significantly improve the performance of the model. Therefore, we use “ResNet1D” and remove the pooling layer in CENTIME. Besides, CENTIME uses statistical features to add information about the overall structure of the traffic. This makes accuracy and F_1 of CENTIME improved a little bit compared with “ResNet1D”.

TABLE IV: Result of the CENTIME Comparing with State-of-the-art Models

Classifier	Accuracy	Precision	Recall	F_1
FlowPic [24]	0.88	-	-	-
CNN1D [8]	-	0.85	0.86	0.86
Deep Packet-CNN [25]	-	0.94	0.93	0.93
Deep Packet-SAE [25]	-	0.92	0.92	0.92
CNN+LSTM [34]	-	0.91	0.91	0.91
SPCaps [22]	-	0.993	0.993	0.993
CENTIME	0.997	0.998	0.997	0.998

To further evaluate the performance of the CENTIME, we compare our framework with other state-of-the-art models. “Table. IV” shows the accuracy of CENTIME is as high as 99.7% on the “ISCX VPN-nonVPN” dataset. In summary, compared with the other six methods, CENTIME achieves the highest overall accuracy, precision, recall and F_1 .

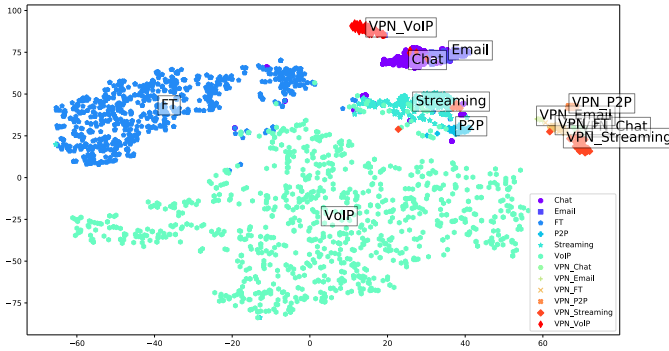


Fig. 5: Visualizations of 12 Types of Encrypted Traffic Data.

E. Visualization of Comprehensive Features

CENTIME can generate comprehensive traffic features based on uniformized traffic files and statistical information. In order to explore the property of the comprehensive features extracted by CENTIME, we use t-SNE [35] to embed the high dimension features into a lower dimension. As shown in “Fig. 5”, the comprehensive traffic features are projected into 2D space. Different colours and different shapes in “Fig. 5” represent different classes of traffic. It can be seen from “Fig. 5” that the same classes of traffic cluster in similar positions, while different classes are far away from each other. For VPN and non-VPN data, VPN data mainly cluster on the right side of the figure, such as “VPN_P2P” and “VPN_Chart”. However, non-VPN data mainly cluster on the left side, such as “FT” and “VoIP”. This shows that comprehensive traffic features can represent the characteristics of different classes of traffic well. Therefore, when CENTIME uses these features to identify encrypted traffic, it can achieve superior performance.

V. CONCLUSION AND FUTURE WORK

Nowadays, encrypted traffic classification is receiving more and more attention from both academic and industrial fields. The current popular methods can be divided into two categories, (1) statistical characteristics-based methods and (2) raw traffic-based methods. However, statistical characteristics-based methods need to design different features for different classification tasks, and using raw traffic-based methods will cause loss of information. Therefore, a new framework called CENTIME is proposed in this paper, which can generate comprehensive traffic features from traffic files and use these features to classify encrypted traffic.

In CENTIME, ResNet is used to extract information directly from uniformized traffic. At the same time, 26 statistics are extracted from traffic files and then encoded by AutoEncoder. Finally, the information extracted by ResNet and encoded by AutoEncoder are merged to form comprehensive traffic features. We use these comprehensive features to classify encrypted traffic, and the results show that the CENTIME yields significant improvements against the other six public methods. Besides, by projecting the features to low dimension, we found that the comprehensive traffic features can represent various traffic types well.

There is still room for improvement in this work. Firstly, the “ISCX VPN-nonVPN” dataset is imbalanced. For example, the amount of traffic “VoIP” is far more than other classes of traffic. Sampling or modifying the loss function can be used to solve the imbalanced problem and improve the performance of models. Secondly, 26 statistics need to be extracted in the CENTIME; it will be more time-consuming than the methods only using deep learning.

Notwithstanding these limitations, this study offers valuable insight into extracting comprehensive traffic features. Additionally, the experiments in this paper have proved that using the 1D convolution layer and removing the pooling layer can achieve better performance on the encrypted traffic classification task. In the future, more work needs to focus on identifying encrypted traffic under the imbalance dataset and making the framework work in real-time.

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